

Variational Hamiltonian Descent

Rimika Jaiswal

*Department of Physics, University of California, Santa Barbara, CA**

Onur Yararbas and Maya Ma

Department of Computer Science, University of California, Santa Barbara, CA

Murphy Yuezhen Niu

*Department of Computer Science, University of California, Santa Barbara, CA and
Google Quantum AI, Santa Barbara, CA[†]*

Preparing highly entangled ground states with shallow variational circuits remains a challenging task for near-term quantum devices. We introduce Variational Hamiltonian Descent (VHD), a class of variational ansätze inspired by Quantum Hamiltonian Descent, in which target evolution is alternated with a kinetic term that is strongest in the early part of the circuit and weaker at later layers. We show that, although coherent target-plus-kinetic dynamics do not furnish a universal convergence theorem in general, the same architecture can nevertheless induce favorable variational landscapes. In a solvable quadratic Gaussian setting, we prove that VHD can yield exact ground state preparation together with globally benign optimization structure at low depth, and admits a simple intrinsic description at arbitrary depth. We then test this idea on the critical transverse field Ising model using matrix product state simulations. Across the shallow depth regime most relevant for current hardware, we find that VHD yields at least $4\times$ larger fidelities than standard VQE without run-to-run variability. Under realistic NISQ constraints where the performance of ADAPT-VQE degrades, we find that VHD can surpass ADAPT fidelities with up to 2 orders of magnitude lower sampling costs. Finally, we deploy classically pre-optimized VHD circuits to prepare the ground state of the 50-qubit TFIM on IBM Heron class processors, combining it with an improved error mitigation workflow to enable a large scale error-mitigated demonstration. These results indicate that embedding a kinetic drive directly into the VQE ansatz architecture provides a practical path to high-fidelity ground-state preparation on NISQ hardware.

I. INTRODUCTION

Preparing many-body ground states on near-term quantum hardware remains a central challenge for quantum simulation. The variational quantum eigensolver (VQE) offers a practical route for pre-fault-tolerant devices by dividing the task between quantum state preparation and classical optimization [1, 2]. A parameterized quantum circuit prepares a trial state on the device, while a classical optimizer updates the circuit parameters using energy estimates obtained from measurements. By replacing long coherent algorithmic sequences with short-depth variational circuits, VQE has emerged as a leading approach for applications ranging from quantum chemistry to strongly correlated lattice models [3–5].

Despite its promise, the practical performance of VQE is often shaped less by the variational principle itself than by the optimization landscape induced by the ansatz. At low depth, physically constrained circuits can produce rugged and strongly nonconvex energy landscapes, with suboptimal local traps and pronounced run-to-run variability [6]. At the opposite extreme, highly expressive deeper circuits can develop barren plateaus, where gradients become exponentially suppressed and optimization

correspondingly difficult [7]. These challenges are further amplified in realistic settings, where finite shot budgets and device noise make reliable navigation of even moderately complicated landscapes increasingly costly.

A prominent strategy for addressing these issues is to adapt the ansatz during optimization. Methods such as ADAPT-VQE iteratively grow the circuit by selecting operators that appear most useful for local energy reduction [8]. In idealized high-precision settings, this can mitigate some of the optimization pathologies of fixed ansätze. However, under realistic shot budgets and hardware noise, the adaptive selection step itself becomes a bottleneck. Estimating gradients across a large operator pool requires many circuit executions at every iteration, and inaccurate gradient estimates under finite sampling can lead to incorrect operator choices, wasted circuit depth, and degraded final fidelity. As a result, high fidelity state preparation becomes difficult precisely in the noisy, hardware-constrained regime where near-term quantum devices operate.

In this work, we pursue a different route by asking whether embedding physically motivated dynamics directly into the ansatz structure can mitigate these optimization and resource bottlenecks. Our starting point is Quantum Hamiltonian Descent (QHD), which recasts optimization as a controlled quantum evolution generated by a target potential together with a kinetic-energy-like term whose influence decays in time [9]. The kinetic term

* rimika@ucsb.edu

[†] murphyniu@ucsb.edu

promotes early exploration of Hilbert space, while its decay slowly biases the dynamics toward low-energy states as the evolution proceeds. This picture suggests a concrete state preparation principle: exploratory kinetic dynamics should be front-loaded early in the circuit, and gradually reduced as the target Hamiltonian takes over.

Before developing this idea, it is important to understand what QHD can and cannot guarantee in the quantum state preparation setting. Along these lines, we establish two no-go theorems for generic coherent target-plus-kinetic dynamics. The first shows that for generic target Hamiltonians, no fixed, state-independent control schedule can guarantee monotonic energy descent. The second shows that even asymptotic convergence is obstructed: a state-independent unitary control law cannot make a unique target state into a genuine global attractor. These results do not undermine the usefulness of the QHD picture for state preparation. Instead, they motivate a variational reframing: can variational optimization provide the state-dependent adaptability that no fixed coherent schedule can achieve?

Motivated by this, we introduce Variational Hamiltonian Descent (VHD), a variational circuit construction that translates the target-plus-kinetic structure suggested by QHD into an ansatz of alternating target Hamiltonian H_T and kinetic K layers. Importantly, while all parameters are optimized freely, their initialization follows the QHD intuition: variational parameters associated with the target Hamiltonian H_T increase with circuit depth, while the parameters associated with the kinetic term are forced to decay. We first validate this design principle in a solvable quadratic setting, where both the target and kinetic generators are quadratic operators and the circuit dynamics remains confined to the manifold of Gaussian states. In this setting, we prove that the VHD ansatz generates a globally well-behaved optimization landscape at any circuit depth: at low depth it enables exact ground-state preparation and global convergence of gradient flow, while at arbitrary depth increasing circuit depth introduces redundancy rather than new intrinsic local minima. Although these results do not extend as rigorous theorems beyond the quadratic sector, they provide a concrete proof of principle that QHD-inspired circuit structure can yield favorable optimization landscapes.

We then benchmark VHD on a demanding many-body problem, the one-dimensional transverse field Ising model (TFIM) at its quantum critical point. It is a stringent test for shallow variational circuits as the spectral gap closes and entanglement grows logarithmically with subsystem size at criticality. Using matrix product state simulations up to $N = 50$ qubits, we demonstrate three distinct advantages. First, the physically motivated initialization eliminates run-to-run variability and achieves up to five times higher fidelity than standard VQE at the same shallow circuit depths. Second, this resource advantage grows with system size: VQE fails to reach moderate fidelity thresholds entirely for $N \gtrsim 25$, while VHD

continues to do so with a native gate count that scales significantly much more favorably. Third, VHD remains robust under both finite shot budgets and realistic device noise, outperforming ADAPT-VQE even when ADAPT is allocated ten times more measurements per function evaluation. Beyond variational energy estimation, we argue that the shallow-depth, moderate-fidelity states produced by VHD can serve as excellent inputs to fault-tolerant algorithms such as quantum phase estimation and QSVT-based ground state preparation, where even a moderate improvements in initial overlap translate into large reductions in downstream algorithmic overhead.

We further validate VHD on IBM’s 156-qubit Heron processors, executing classically pre-optimized circuits for the critical TFIM using $N = 14, 30$, and 50 qubits, all at circuit depth $p = 3$. Combined with the MoSAIC error mitigation protocol [10], we show that we can recover energies close to ideal tensor-network values for all three system sizes. The fact that the same circuit depth suffices across a fourfold increase in system size indicates that the favorable resource scaling seen in simulations persists in realistic hardware settings.

The rest of the paper is organized as follows. In Sec. II we develop the theoretical foundations for VHD, including the no-go theorems, the ansatz derivation and the Gaussian proof of principle. In Sec. III we present numerical results organized around the three advantages described above, comparing VHD against standard VQE and ADAPT-VQE in noiseless, shot-limited, and hardware-noise settings. Sec. IIID details the hardware results. In Sec. IV we discuss the implications, limitations, and future directions. Detailed proofs of the various ansatz, no-go theorems, and the Gaussian landscape analysis are collected in the Appendix.

II. THEORETICAL RESULTS AND MOTIVATION FOR VHD

We now develop the theoretical foundations of the Variational Hamiltonian Descent (VHD) approach. We begin in Sec. II A by analyzing the continuous time Quantum Hamiltonian Descent picture and identifying two fundamental obstructions that prevent it from serving as a literal descent algorithm for quantum state preparation. We then show in Sec. II B how this picture, despite these limitations, motivates a physically structured variational circuit ansatz. In Sec. II C we validate the resulting architecture in a solvable quadratic setting, demonstrating that it produces a provably benign optimization landscape already at low circuit depths. Finally, in Sec. II D we turn to the transverse field Ising model and show how the VHD design principle can be concretely implemented for a challenging, strongly correlated benchmark.

A. Limitations of the continuous time picture

In the Quantum Hamiltonian Descent framework, one approaches an optimization problem by studying a continuous time evolution generated by a Hamiltonian consisting of two competing pieces: a potential term that mimics the objective function's landscape, and a kinetic term that drives motion across that landscape. The basic intuition is simple. The kinetic contribution is strongest early in the evolution, when broad exploration of parameter space is desired, and then gradually weakens so that the dynamics becomes increasingly dominated by the objective itself, guiding the system toward the global minimum. In this way, the kinetic term promotes early exploration of the landscape, while the target term gradually biases the dynamics toward low energy states.

If we borrow this template for quantum state preparation, the natural identification is to take the objective term to be the target Hamiltonian H_T whose ground state we wish to prepare. This leads to the continuous time control Hamiltonian

$$H(t) = a(t)H_T + b(t)K, \quad (1)$$

where K is a kinetic energy-like term and $a(t)$, $b(t)$ are real-valued control functions. Crucially, K should be chosen so that it can drive transitions between different eigenstates of H_T , ensuring that the evolution does not remain confined to a single sector of Hilbert space. In practice, this can be a difficult condition to engineer or verify in full generality, but at a minimum the kinetic term should not commute with the target Hamiltonian,

$$[K, H_T] \neq 0,$$

so that it can induce nontrivial dynamics.

At first sight, one might hope that such a target-plus-kinetic evolution would itself provide a coherent descent mechanism toward the ground state of H_T . However, for generic target Hamiltonians this hope fails, and two distinct obstructions can be identified. Together, they explain why the QHD template cannot be taken literally to have convergence guarantees for state preparation, and why a different strategy is needed.

The first obstruction is local in time. For general target-plus-kinetic dynamics, the instantaneous rate of change of the target energy is

$$\dot{E}(t) = i b(t) \langle \psi_t | [K, H_T] | \psi_t \rangle,$$

which depends on the state $|\psi_t\rangle$ through the commutator expectation value $\langle [K, H_T] \rangle$. Since this quantity can be positive for some states and negative for others, the same control schedule can simultaneously push certain states downhill in energy while pushing others uphill. This observation can be sharpened into a precise no-go statement.

Theorem 1. *Let $|\psi_t\rangle$ evolve under*

$$i\partial_t |\psi_t\rangle = (a(t)H_T + b(t)K) |\psi_t\rangle,$$

where $[K, H_T] \neq 0$, and let $E(t) = \langle \psi_t | H_T | \psi_t \rangle$ be the instantaneous target energy. Then, no fixed, state-independent control functions $a(t)$ and $b(t)$ can make the target energy $E(t)$, or any strictly monotone function thereof, a universal Lyapunov function for convergence to the ground state of H_T .

In other words, there is no state-independent control law that guarantees monotonic energy descent for all initial states simultaneously. Any schedule that works well for one initial condition may fail, or even increase the energy, for another.

The second obstruction is global. Even if one abandons the requirement of monotonic descent and asks only for asymptotic convergence, a state-independent unitary evolution cannot make a unique target ray into a genuine global attractor on any nontrivial set of initial conditions. The reason is simply that unitary dynamics preserves inner products: two distinct initial states cannot both converge to the same final pure state unless their overlap was already equal to one. We note that this observation is also consistent with the broader closed quantum control literature, where purely Hamiltonian dynamics is confined to isospectral unitary orbits and global asymptotic stabilization is known to be obstructed in closed systems [11]. Detailed statements and proofs of both obstructions are collected in the Appendix.

These two no-go statements clarify the role of the QHD picture in the present work. Its contribution is to suggest a physically structured family of generators that can be promoted into a variational circuit ansatz. The resulting ansatz then acquires the flexibility needed for practical state preparation through variational optimization.

B. From continuous control to a variational ansatz

To construct the VHD ansatz, we first partition the total evolution time T into p slices of equal width,

$$\Delta t = T/p, \quad t_j = j \Delta t, \quad j = 0, \dots, p.$$

For each slice, the cumulative effect of the target and kinetic terms is captured by the integrated angles

$$\theta_j := \int_{t_j}^{t_{j+1}} a(t) dt, \quad \phi_j := \int_{t_j}^{t_{j+1}} b(t) dt. \quad (2)$$

A first order Trotterization of the continuous control evolution then takes the form

$$U_{\text{ctrl}}^{(p)} \approx \prod_{j=1}^p e^{-i\theta_j H_T} e^{-i\phi_j K}. \quad (3)$$

This discrete circuit retains the original QHD intuition: early layers carry larger kinetic angles and therefore stronger exploratory action, while later layers can become progressively dominated by the target Hamiltonian.

The key step is now to stop treating the slice angles $\{\theta_j, \phi_j\}$ as fixed by a prescribed control schedule, and instead promote them to free variational parameters. This defines the Variational Hamiltonian Descent ansatz,

$$|\psi_{\text{VHD}}(\boldsymbol{\theta}, \boldsymbol{\phi})\rangle = \prod_{j=1}^p e^{-i\theta_j H_T} e^{-i\phi_j K} |\psi_0\rangle, \quad (4)$$

where $|\psi_0\rangle$ is an easy-to-prepare reference state. The variational parameters are then optimized by minimizing the target energy in the trial state,

$$E(\boldsymbol{\theta}, \boldsymbol{\phi}) = \langle \psi_{\text{VHD}}(\boldsymbol{\theta}, \boldsymbol{\phi}) | H_T | \psi_{\text{VHD}}(\boldsymbol{\theta}, \boldsymbol{\phi}) \rangle, \quad (5)$$

which is the standard variational objective. The continuous time picture thus provides the architecture for the ansatz, while the variational optimization provides the adaptability needed for practical state preparation.

C. Solvable quadratic case: proof of principle

We now show, in a setting simple enough to admit a fully analytic treatment, that the VHD architecture can indeed produce a provably favorable optimization landscape. The key ideas are developed here; full statements and proofs are given in the appendix.

Consider the harmonic oscillator target Hamiltonian

$$H_\omega = \frac{1}{2}(P^2 + \omega^2 X^2), \quad K = \frac{1}{2}(XP + PX),$$

where K is the generator of squeezing. We choose this particular kinetic operator because it can systematically change the width of the state in phase space. Specifically, starting from the vacuum of

$$H_1 = \frac{1}{2}(P^2 + X^2),$$

the action of $e^{-i\phi K}$ stretches one quadrature and compresses the other,

$$X \mapsto e^\phi X, \quad P \mapsto e^{-\phi} P,$$

moving the state along the manifold of centered pure Gaussian states.

This makes the depth one VHD ansatz immediately transparent. For

$$|\psi(\theta, \phi)\rangle = e^{-i\theta H_\omega} e^{-i\phi K} |0\rangle,$$

the final target layer $e^{-i\theta H_\omega}$ does not affect the variational energy, since H_ω commutes with its own exponential. The only nontrivial parameter is therefore the squeezing angle ϕ . Evaluating the energy thus gives

$$E(\phi) = \frac{1}{4}(e^{-2\phi} + \omega^2 e^{2\phi}),$$

reflecting the competition between overcompressing and oversqueezing the two quadratures. This function is strictly convex, with a unique global minimum at

$$\phi_* = -\frac{1}{2} \log \omega,$$

at which point it exactly reproduces the ground state energy $\omega/2$. Notably, already at depth one the VHD structure reduces the state preparation problem to a one-dimensional optimization with no spurious local minima.

The depth two case is more revealing. One might worry that inserting an intermediate target evolution layer would genuinely complicate the landscape. Instead, the opposite occurs. For

$$|\psi(\phi_1, \phi_2)\rangle = e^{-i\theta_2 H_\omega} e^{-i\phi_2 K} e^{-i\theta_1 H_\omega} e^{-i\phi_1 K} |0\rangle,$$

the energy can still be written exactly, and after the change of variables

$$u = \phi_1 + \phi_2, \quad v = \phi_1 - \phi_2,$$

it separates into two weighted copies of the depth one profile,

$$E(\phi_1, \phi_2) = \cos^2(\omega\theta_1) E_{\text{single}}(u) + \sin^2(\omega\theta_1) E_{\text{single}}(v),$$

where $E_{\text{single}}(x) = \frac{1}{4}(\omega^2 e^{2x} + e^{-2x})$. The landscape therefore does not acquire new intrinsic traps at depth two: it remains globally strongly convex, and the minimizer can always reach the exact ground state. Physically, the optimization still reduces to choosing the correct effective squeezing, now resolved along two simple linear combinations of the circuit parameters.

This pattern is not an accident of low depth. The deeper reason is that both H_ω and K are quadratic generators, so any finite depth VHD circuit remains within the same low dimensional manifold of centered pure Gaussian states. On that manifold, every state is characterized by an overall phase space orientation and a single effective squeezing magnitude r , and the target energy depends only on this radial coordinate,

$$E = \frac{\omega}{2} \cosh(2r). \quad (6)$$

The orientation drops out entirely.

The Gaussian prototype therefore has a unique intrinsic minimum and no spurious intrinsic traps. Increasing circuit depth mainly introduces redundant parameterizations at the level of the state manifold rather than new intrinsic minima (Theorem 7). This illustrates how the VHD architecture, despite lacking a universal convergence theorem, can still induce remarkably benign optimization structure within a structured sector of Hilbert space.

One may then ask whether the continuous time QHD dynamics itself converges within the Gaussian sector, or whether the mechanism of the no-go theorems of Sec. II A is already visible here. In fact, the no-go phenomenon persists even in this solvable case. On the manifold of centered pure Gaussian states, the target energy evolves according to

$$\dot{E} = \omega b(t) \sinh(2r) \cos(2\alpha),$$

where $r \geq 0$ is the effective squeezing magnitude and α is the orientation angle. This expression takes both positive and negative values across the manifold, confirming

that no fixed, state-independent schedule can guarantee monotonic energy descent, even within the Gaussian sector.

At the same time, this does not contradict the positive state preparation result above. The no-go statement applies uniformly over all initial states, whereas the Gaussian example concerns a particular reference state, namely the vacuum of H_1 , from which the ground state of H_ω lies on the same squeezed orbit and can therefore be reached by a suitably chosen coherent schedule. VHD just helps discover the optimal path through variational optimization.

D. VHD for the transverse field Ising model

The exact solubility of the Gaussian prototype is highly special, relying on the fact that both the target and kinetic generators are quadratic. In the settings of practical interest, namely the ground state preparation of interacting many-body Hamiltonians, the dynamics inevitably leaves the tractable Gaussian sector and these rigorous geometric guarantees dissolve. It is therefore important to ask whether the VHD design principle remains effective in more realistic, strongly correlated settings.

To address this question, we turn to the one-dimensional transverse field Ising model (TFIM),

$$H_{\text{TFIM}} = H_{ZZ} + H_X, \quad (7)$$

with

$$H_{ZZ} = -J \sum_i Z_i Z_{i+1}, \quad H_X = -h \sum_i X_i, \quad (8)$$

where J is the nearest-neighbor Ising coupling and h is the transverse field strength. We focus specifically on ground state preparation at the quantum critical point $J = h$. This regime is particularly challenging for both classical and quantum algorithms: the spectral gap closes in the thermodynamic limit [12], and the entanglement entropy grows logarithmically with subsystem size [13]. As a result, shallow variational circuits often struggle to represent the critical ground state accurately, while adaptive strategies can become increasingly expensive under finite sampling and device noise. The critical TFIM therefore serves as a great benchmark for testing whether VHD remains effective beyond the analytically tractable Gaussian setting.

1. Choice of the kinetic drive

As discussed in Sec. II A, the kinetic operator K should be chosen so that it generates dynamics genuinely distinct from the native target evolution. At a minimum, it should not commute with H_T , so that it can move the state across different eigenspaces of the Hamiltonian and induce nontrivial transitions. For our TFIM benchmark,

we also impose a practical requirement: the kinetic term should be local and hardware efficient, so that the resulting protocol remains compatible with near-term quantum devices. With these considerations in mind, we choose the uniform transverse kinetic drive

$$K = \sum_i Y_i. \quad (9)$$

This choice is local, straightforward to implement, and noncommuting with both H_{ZZ} and H_X . It therefore introduces dynamics that are complementary to the target evolution while remaining experimentally realistic.

A single VHD layer for the TFIM is thus given by

$$U_m(\theta_m^{ZZ}, \theta_m^X, \Phi_m) = e^{-i\theta_m^{ZZ} H_{ZZ}} e^{-i\theta_m^X H_X} e^{-i\Phi_m K}, \quad (10)$$

and the full depth p VHD ansatz reads

$$|\psi_{\text{VHD}}(\theta^{ZZ}, \theta^X, \Phi)\rangle = \prod_{m=1}^p U_m(\theta_m^{ZZ}, \theta_m^X, \Phi_m) |\psi_0\rangle, \quad (11)$$

where $|\psi_0\rangle$ is a simple, easy-to-prepare reference state to be specified below. Each layer thus consists of a target evolution block, split into its H_{ZZ} and H_X components, followed by a kinetic rotation generated by K .

2. Practical TFIM-specific VHD parameterization

In its most general form, Eq. (11) treats all target and kinetic angles as independent variational parameters. For the TFIM numerics presented in this paper, however, we adopt a reduced parameterization that directly encodes the qualitative QHD intuition: the kinetic contribution should be strongest in the early layers, when broad exploration is most valuable, and should weaken as the optimization progresses. Concretely, we optimize the target angles $\{\theta_m^{ZZ}, \theta_m^X\}$ freely, while fixing the kinetic angles to a monotonically decaying schedule,

$$\Phi_m = A e^{-\zeta(m-1)}. \quad (12)$$

Here A sets the initial kinetic strength and $\zeta > 0$ controls the rate of decay with layer index. The first layer, therefore, carries the strongest kinetic action, and later layers are progressively dominated by the target Hamiltonian. This implements, in discrete form, the same exploration-to-refinement pattern that motivates the original QHD picture.

With this reduced parameterization, the TFIM VHD ansatz takes the form

$$|\psi_{\text{VHD}}(\theta^{ZZ}, \theta^X; A, \zeta)\rangle = \prod_{m=1}^p e^{-i\theta_m^{ZZ} H_{ZZ}} e^{-i\theta_m^X H_X} e^{-i\Phi_m(A, \zeta) K} |+\rangle^{\otimes N}. \quad (13)$$

We choose $|+\rangle^{\otimes N}$ as the reference state because it is easy to prepare and coincides with the exact ground state in the limit $h \gg J$, which makes it a physically motivated starting point for the variational optimization. For the variational parameters associated with the target evolution, the initial values are taken to increase monotonically with layer index according to a cubic ramp,

$$(\theta_m^{ZZ})^{(0)} \propto t_m^3, \quad (\theta_m^X)^{(0)} \propto t_m^3. \quad (14)$$

This initialization is motivated by the QHD intuition that the relative influence of the target Hamiltonian should grow over the course of the evolution. For the translation-invariant TFIM instances studied here, we find empirically that values near $A \approx \pi/2$ and $\zeta \approx 2.0$ perform consistently well across the range of system sizes and circuit depths considered, so a coarse outer-loop search over these kinetic hyperparameters is sufficient in practice.

The VHD ansatz construction thus preserves the physically motivated alternating target-plus-kinetic structure inherited from the continuous time QHD template. The variational optimization supplies the flexibility needed to adapt this architecture to the target Hamiltonian of interest.

III. RESULTS FOR TFIM

We now present numerical evidence for the advantages of the VHD ansatz over standard VQE and adaptive baselines. Our benchmarks focus on the critical transverse field Ising model at $J = h$, the most challenging regime for shallow variational circuits, as discussed in Sec. IID. We organize our results around three distinct advantages of the VHD approach. We first show in Sec. III A that the physically motivated initialization of VHD eliminates run-to-run variability and yields substantially higher fidelities than standard VQE at fixed shallow depth. We then demonstrate in Sec. III B that this resource advantage grows with system size, with VHD reaching target fidelities at significantly lower native gate counts than VQE, and VQE failing to reach those targets altogether at larger system sizes. Finally, in Sec. III C we show that VHD remains robust under both finite shot budgets and realistic device noise, outperforming ADAPT-VQE even when ADAPT is allocated up to ten times more measurement shots. Throughout, we report energies and fidelities relative to the exact ground state obtained by exact diagonalization for small systems and DMRG for system sizes up to $N = 50$.

A. Advantage I: Physically motivated initialization

A central practical difficulty with standard VQE is sensitivity to initialization. Generic random parameter initializations can land in poor local minima or flat regions of the landscape, leading to strong run-to-run variability

and near-zero fidelities even at moderate circuit depths. As discussed in Sec. IID, the VHD ansatz avoids this by embedding a physically motivated initialization directly into its structure: the kinetic decay schedule and the cubic ramp for the target angles both encode the QHD intuition that exploration should dominate early and refinement should dominate late. We demonstrate the practical consequences of this structural bias in Fig. 1.

Figure 1 shows the converged energy (left) and ground state fidelity (right) as a function of circuit depth for a system of $N = 24$ qubits, comparing VHD (orange) and standard VQE (blue) against the exact ground state energy (red dashed line). We define the fidelity as

$$\mathcal{F} = |\langle \psi_{\text{GS}} | \psi(\boldsymbol{\theta}^*) \rangle|^2, \quad (15)$$

where $|\psi_{\text{GS}}\rangle$ is the exact ground state obtained from exact diagonalization or DMRG, and $|\psi(\boldsymbol{\theta}^*)\rangle$ is the variationally optimized trial state. For standard VQE, we perform 10 independent runs from random parameter initializations and report the median fidelity, with error bars indicating the 25th and 75th percentiles. We use the median instead of the mean because the distribution of VQE outcomes is strongly skewed: the vast majority of random initializations converge to poor solutions with near-zero fidelity, while occasionally a favorable initialization produces a significantly better result. In this setting, the mean is pulled upward by rare good runs and does not faithfully represent typical performance, while symmetric error bars around the mean give a misleading impression of the spread. The median and interquartile range together give a more honest picture of what a practitioner should expect on a given run.

Across all depths from $p = 2$ to $p = 10$, VHD consistently achieves lower energies and higher fidelities than VQE. The fidelity comparison is particularly striking. Standard VQE produces near-zero median fidelity at essentially all shallow depths, with the wide interquartile intervals at depth $p = 9-10$ confirming severe run-to-run variability: the optimizer finds very different solutions on independent runs, and even the higher quartile rarely reaches meaningful fidelity. VHD, by contrast, produces consistent fidelities across runs, rising from approximately 0.47 at depth two to around 0.67 at depth six. In the low-depth regime, VHD achieves up to five times higher fidelity than VQE at the same circuit depth.

We note that neither method reaches unit fidelity within this depth range, which is expected, since at criticality the ground-state entanglement grows logarithmically with system size, and the circuits considered here are too shallow to achieve exact ground-state preparation. The important observation to make is that VHD reliably reaches fidelities ≥ 0.5 at very shallow depths where standard VQE simply fails to deliver.

This shallow-depth, moderate-fidelity capability matters beyond variational energy estimation. Preparing a good approximate ground state at low depths is an important primitive for several fault-tolerant quantum algorithms whose cost depends sensitively on the initial

TABLE I. **Benchmarking setup for all methods.** Summary of ansatz families, initialization strategies, and simulation conditions used in all numerical comparisons. All fixed-ansatz simulations use COBYLA as the optimizer, while the adaptive algorithm (ADAPT-VQE) uses BFGS, following the implementation in Reference [14].

Method	Ansatz: each layer	Initialization	Conditions tested
VQE (problem-specific)	Target Hamiltonian: $H_X + H_{ZZ}$	Random (10 runs)	Ideal; 1k shots; Heron r2 noise
VHD	Target Hamiltonian + Kinetic term ($\sum_i Y_i$)	QHD-inspired (growing target, decaying kinetic schedule)	Ideal; 1k shots; Heron r2 noise
VQE (hardware-efficient)	HEA: Single-qubit rotations + CZ	Random (10 runs)	Ideal; 1k shots; Heron r2 noise
VHD (hardware-efficient)	HEA + Kinetic term ($\sum_i Y_i$)	QHD-inspired (growing target, decaying kinetic schedule)	Ideal; 1k shots; Heron r2 noise
ADAPT-VQE	Adaptive (operator pool: 1- and 2-qubit Paulis)	TETRIS-constructed blocks + ADAPT greedy selection	1k–100k shots; Heron r2 noise

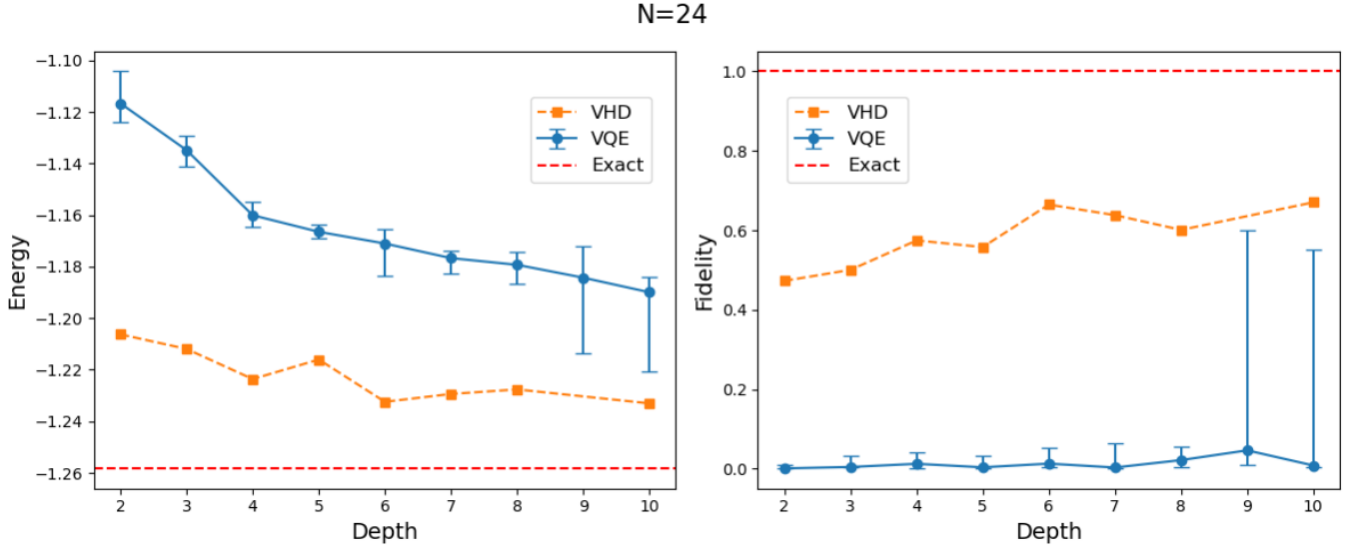


FIG. 1. **Physically motivated initialization eliminates run-to-run variability.** Converged energy (left) and ground state fidelity (right) as a function of circuit depth p for the critical TFIM at $N = 24$, comparing VHD (orange) and standard VQE (blue). Error bars indicate run-to-run variability over independent initializations. The red dashed line marks the exact ground state energy and unit fidelity, respectively. VHD achieves up to five times higher fidelity than VQE at the same depth, while eliminating run-to-run variability throughout the depth range shown.

overlap with the true ground state. In quantum phase estimation [15], the probability of successfully projecting onto the ground state scales as $|\langle\psi_{\text{GS}}|\psi_{\text{init}}\rangle|^2$, so a poor initial state directly multiplies the number of repetitions required. More generally, quantum signal processing and QSVT-based ground-state preparation methods [16, 17] implement polynomial filters that amplify the ground-state component, with both runtime and success probability scaling inversely with the initial overlap. Consequently, in both settings, even moderate improvements in initial state fidelity can lead to substantial reductions in the fault-tolerant overhead. From this perspective, VHD circuits, which reliably achieve fidelities ≥ 0.5 at depths as low as $p = 2$ –3, can serve as great initial state-preparation subroutines for early fault-tolerant devices, as VHD circuits can provide improved initial overlap without incurring a large preparation overhead.

B. Advantage II: Growing resource advantage with system size

Having established that VHD outperforms VQE at fixed system size and fixed depth, we now examine how this advantage scales with system size. The relevant question for near-term applications is not only whether VHD achieves higher fidelity at a given depth, but whether the total hardware resources required to reach a target fidelity grow more favorably with N . To address this, we measure the minimum native gate count required for each method to reach a prescribed target fidelity, and track how this resource cost scales with system size up to $N = 50$.

Figure 2 displays these resource requirements for two target fidelity thresholds. The left panel shows results for a target fidelity of 0.3, a modest threshold that both methods can reach across the full range of system sizes

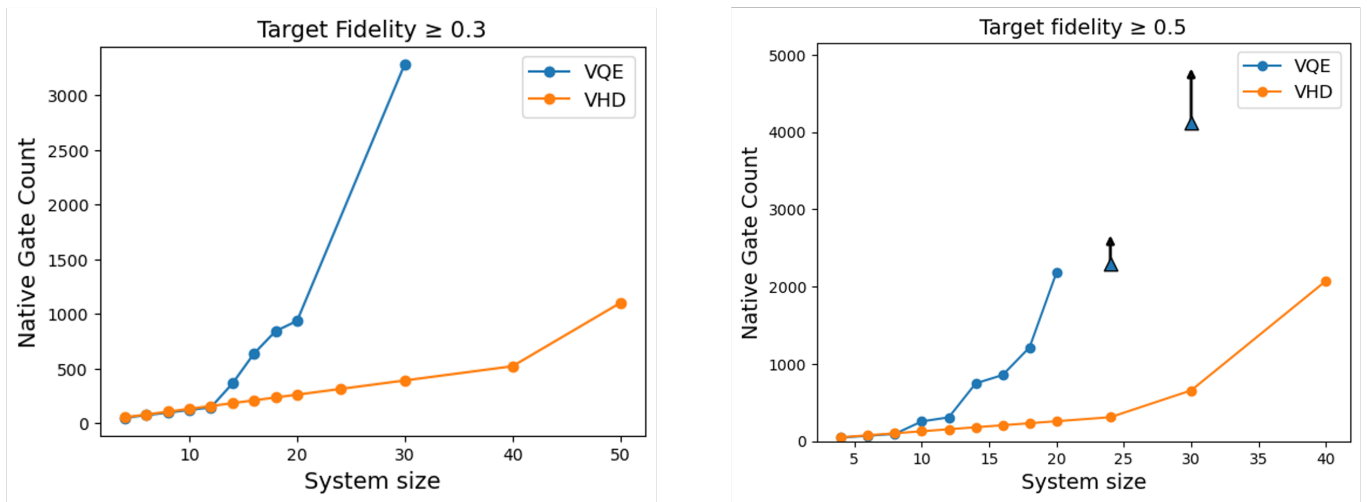


FIG. 2. **Resource advantage of VHD grows with system size.** Minimum native gate count required to reach a target fidelity of ≥ 0.3 (left) and ≥ 0.5 (right) as a function of system size N , comparing VHD (orange) and standard VQE (blue). Upward arrows in the right panel indicate that VQE failed to reach the target fidelity within the circuit depths explored. VHD reaches both thresholds with substantially fewer gates at all system sizes, and the resource gap grows with N .

considered. The right panel shows results for a more demanding threshold of 0.5. In both cases, VHD (orange) requires substantially fewer native gates than VQE (blue), and this advantage grows with N . For the lower threshold, VQE requires roughly three times more gates than VHD at $N = 40$, with the gap growing visibly as the system size increases. For the higher threshold, the picture is more dramatic: standard VQE fails to reach fidelity 0.5 for $N \geq 25$ within the circuit depths we explored, as indicated by the upward arrows in the right panel, while VHD continues to reach the target across all system sizes up to $N = 40$ with a native gate count of approximately 2000. This clearly illustrates that the resource advantage of VHD over standard VQE increases with system size.

We attribute this growing advantage to the physics-structured inductive bias of the VHD ansatz. As the system size increases at the critical point, the ground state becomes increasingly entangled and harder to reach from a generic initialization. The QHD-inspired kinetic drive of VHD systematically pre-oriens the circuit toward the relevant region of Hilbert space, reducing the effective optimization depth needed to reach a given fidelity. Standard VQE, lacking this structural prior, must compensate through deeper circuits and more optimization iterations, a strategy that becomes increasingly costly as both the landscape complexity and the system size grow.

C. Advantage III: Robustness under shot noise and device noise

The advantages demonstrated above were obtained under idealized, noiseless simulation conditions. For near-term applications, however, two additional sources of

degradation are unavoidable: finite measurement shot budgets, which introduce statistical noise into gradient estimates, and hardware noise from gate imperfections and decoherence on physical quantum devices. We now show that VHD retains its advantage under both of these realistic conditions, and that the fixed, physics-structured ansatz is particularly well suited to the constrained resource regime where adaptive methods struggle most.

1. Finite shot budgets

We first consider the effect of finite sampling. Adaptive methods such as ADAPT-VQE are particularly sensitive to shot noise because they require accurate gradient estimates across a large operator pool at every selection step. Inaccurate estimates lead to incorrect operator choices, wasted circuit depth, and degraded final fidelity, a failure mode that worsens as the shot budget is reduced. To ensure a fair comparison, all ADAPT-VQE simulations in this paper are implemented using the state-of-the-art CEO-ADAPT codebase [14], the current most resource-efficient implementation of the adaptive algorithm.

VHD, being a fixed-structure ansatz, avoids this vulnerability: once the kinetic schedule is set, only the target angles require optimization, and the physics-structured initialization reduces the number of optimization iterations needed to reach a given fidelity. Figure 3 illustrates this directly. We plot fidelity as a function of total native gate count for VHD with 1000 shots per energy evaluation (orange), standard VQE with 1000 shots (blue), and ADAPT-VQE with 5000 and 10000 shots (light and dark green, respectively). VHD reaches a fidelity of approximately 0.88 at a gate count of around 350, while ADAPT-

VQE with ten times the shot budget reaches only approximately 0.57 at comparable gate counts, and plateaus well below the VHD curve. Standard VQE with the same 1000-shot budget performs better than ADAPT in this resource-constrained regime, reaching around 0.65, but still falls significantly short of VHD. These results confirm that VHD outperforms ADAPT-VQE even when ADAPT is allocated up to ten times more measurements per cost function evaluation.

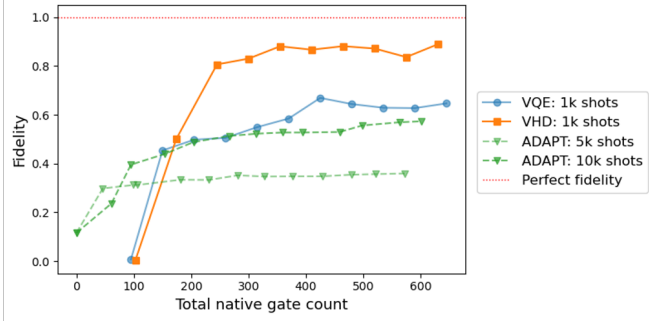


FIG. 3. **VHD outperforms ADAPT-VQE with ten times fewer shots.** Fidelity as a function of total native gate count for $N = 8$ qubits showing VHD with 1000 shots (orange), VQE with 1000 shots (blue), and ADAPT-VQE with 5000 and 10000 shots (light and dark green, respectively). VHD reaches fidelity ≈ 0.88 while ADAPT with ten times the shot budget plateaus at ≈ 0.57 .

2. Realistic device noise

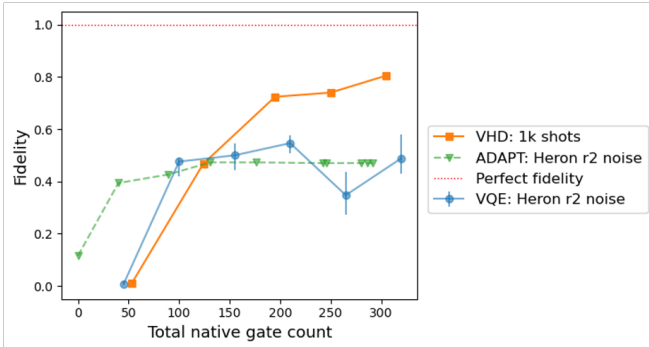


FIG. 4. **VHD is robust under realistic device noise.** Fidelity versus total native gate count under IBM Heron r2 noise model for $N = 8$ qubits showing VHD (orange), VQE (blue), and ADAPT-VQE (green dashed). VHD reaches fidelity ≈ 0.80 , substantially outperforming both VQE (≈ 0.55 , with large run-to-run variability) and ADAPT (≈ 0.48 , plateaued) at comparable gate counts.

We next test VHD under a realistic hardware noise model. Specifically, we simulate the noise characteristics of the IBM Heron r2 processor, which includes two-qubit gate errors, readout errors, and idle decoherence. Under

these conditions, longer circuits accumulate more error, so the effective fidelity can decrease with circuit depth beyond a certain point, even if the noiseless fidelity would continue to improve. This penalizes methods that require deeper circuits to make meaningful progress, and rewards methods that achieve high fidelity at low depth.

Figure 4 shows fidelity versus total native gate count, for all three methods subjected to the Heron r2 noise model, comparing VHD with 1000 shots (orange), VQE under the noise model (blue), and ADAPT-VQE under the noise model (green dashed). VHD rises steadily to a fidelity of approximately 0.80 at a gate count of around 320, while VQE under noise reaches only approximately 0.55 with large error bars indicating significant run-to-run instability, and ADAPT under noise plateaus at approximately 0.48 and fails to improve further. The fact that VHD substantially outperforms both noisy VQE and noisy ADAPT highlights the primary advantage of the physics-structured shallow circuit: by reaching high fidelity at low gate depth, VHD exposes the state to less accumulated hardware noise than either of the competing methods.

D. Hardware implementation

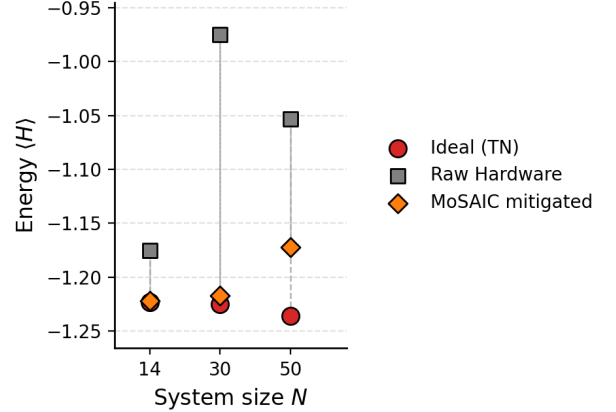


FIG. 5. **VHD ground state energies on IBM quantum hardware.** Ground state energies per site for the critical TFIM at $N = 14, 30$, and 50 qubits, comparing ideal tensor network results (red circles), raw hardware outputs (grey squares), and error-mitigated results using the MoSAIC protocol (orange diamonds). Error mitigation substantially closes the gap between raw hardware and ideal across all system sizes, demonstrating the practical viability of VHD-prepared states on current quantum processors.

To assess the practical viability of VHD on current quantum hardware, we deploy classically optimized VHD circuits directly on IBM's Heron r2 quantum processors. We first obtain the optimal variational parameters using large scale tensor network simulations. We then transpile and execute the optimized circuit on hardware. This ap-

proach is a sensible attractive strategy for near-term devices, where the shot overhead of on-device variational optimization can be prohibitive and classical simulation remains tractable for the system sizes considered.

The specific circuits and depth choices are detailed in the accompanying code repository [18]. Briefly, for the $N = 14$ qubit experiment, we use the hardware-efficient ansatz of Eq. (A3) at circuit depth $p = 3$, while for $N = 30$ and $N = 50$ we use the problem-tailored target-plus-kinetic ansatz of Eq. (13) at the same depth $p = 3$. Notably, the same depth circuit suffices to prepare good-quality ground states across system sizes from 14 to 50. This is a direct consequence of the favorable depth scaling of VHD established in Sec. III B: the physics-structured initialization pre-oriens the circuit toward the ground state manifold with an efficiency that does not degrade rapidly with N , so that the required circuit depth grows far more slowly than for generic variational ansätze. On hardware, this translates directly into lower accumulated noise at larger system sizes, since the total gate count does not grow proportionally with N .

Raw hardware outputs, shown as grey squares in Fig. 5, exhibit significant deviations from the ideal tensor network energies (red circles) at all system sizes, as expected from two-qubit gate errors, readout noise, and idle decoherence accumulating over the circuit. To recover accurate energy estimates, we apply MoSAIC [10], an improved probabilistic error cancellation (PEC) protocol that reduces the sampling overhead of standard PEC by exploiting the structure of the noise channel. The resulting mitigated energies (orange diamonds) substantially close the gap to the ideal values across all three system sizes. At $N = 14$ and $N = 30$, the mitigated energies lie very close to the tensor network reference, while at $N = 50$ a small deviation remains.

Taken together, these results demonstrate that shallow VHD circuits can produce energy estimates that are both meaningful and correctable on current IBM processors up to $N = 50$ qubits. The combination of shallow circuit depth from physics-structured ansatz design and effective error mitigation from MoSAIC provides a practical path toward large-scale ground state preparation on near-term devices.

IV. DISCUSSION AND OUTLOOK

To summarize, we showed that by embedding the alternating target-plus-kinetic structure of Quantum Hamiltonian Descent directly into the circuit ansatz, VHD encodes a physically motivated inductive bias that addresses the two principal failure modes of near-term variational algorithms simultaneously: the run-to-run instability of generic VQE, and the measurement overhead that makes adaptive methods such as ADAPT-VQE fragile under realistic resource constraints. These two properties compound favorably with system size. At small N , the advantage of VHD over VQE is primarily in reli-

ability; as N grows toward and beyond the critical point, standard VQE begins to fail entirely at reaching moderate fidelity thresholds within practical circuit depths, while VHD continues to make consistent progress.

This picture is consistent with, but goes beyond, the provable Gaussian results of Sec. II C. In the quadratic setting, the benign structure of VHD follows from the fact that the circuit remains within the manifold of centered pure Gaussian states, on which the target energy depends only on a single intrinsic squeezing coordinate. At low depth this yields explicit convexity results in the circuit parameters, while at arbitrary depth it implies that additional layers mainly introduce redundant parameterizations and crucially no new intrinsic minima. In the interacting TFIM, these geometric guarantees dissolve, and we cannot at present prove that the VHD landscape is trap-free in general. What the numerical results demonstrate is that the physics-structured initialization is sufficiently well-aligned with the target ground state that the optimizer consistently finds good solutions without the repeated random restarts that standard VQE requires. Whether this empirical benignness reflects an approximate version of the Gaussian landscape structure, or a different mechanism altogether, is the most important open theoretical question raised by this work.

Several caveats accompany the results presented here and each suggests a clear direction for future work.

a. Model specificity and kinetic drive selection. The kinetic drive $K = \sum_i Y_i$ and the decaying schedule $\Phi_m = Ae^{-\zeta(m-1)}$ were chosen based on physical intuition and validated empirically for the TFIM. For other target Hamiltonians, the appropriate kinetic drive may be quite different, and identifying principled selection criteria beyond the minimal noncommutativity requirement remains an open problem. One promising direction, motivated by quantum optimal control, is to treat K itself as a variational object, optimizing over a family of candidate kinetic operators subject to locality and hardware constraints. Similarly, the hyperparameters $A \approx \pi/2$ and $\zeta \approx 2.0$ were found through a coarse search; a more systematic theory of kinetic schedule design would strengthen the method's applicability to new settings.

b. Theoretical landscape analysis. The key open theoretical question is whether the benign empirical optimization landscape of VHD reflects a more general principle that can be made precise beyond the Gaussian sector. Two particularly relevant directions are i) perturbative analysis around the solvable quadratic case, and ii) a direct study of gradient scaling with system size to assess whether the physics-structured initialization provides systematic protection against barren plateaus [7, 19]. Such results would provide a much firmer foundation for the VHD design principle.

c. Extensions. The results in this paper focus on a one-dimensional spin model simulated classically. Extending VHD to fermionic systems, higher-dimensional lattices, and molecular Hamiltonians relevant for quantum chemistry are important directions with immediate

practical relevance.

V. ACKNOWLEDGMENTS

R.J. was supported by the UCSB NSF Quantum Foundry. This research used resources of the National Energy Research Scientific Computing Center (NERSC), a Department of Energy Office of Science User Facility

under Contract No. DE-AC02-05CH11231 using NERSC award DDR-ERCAP0021493. This research is also supported by the U.S. National Science Foundation grant CCF-2441912(CAREER), Air Force Office of Scientific Research under award number FA9550-25-1-0146, and the U.S. Department of Energy, Office of Advanced Scientific Computing Research under Award Number DE-SC0025430.

-
- [1] A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, and J. L. O'Brien, A variational eigenvalue solver on a photonic quantum processor, *Nature Communications* **5**, 4213 (2014).
 - [2] J. R. McClean, J. Romero, R. Babbush, and A. Aspuru-Guzik, The theory of variational hybrid quantum-classical algorithms, *New Journal of Physics* **18**, 023023 (2016).
 - [3] A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, and J. M. Gambetta, Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets, *Nature* **549**, 242 (2017).
 - [4] D. Wecker, M. B. Hastings, and M. Troyer, Progress towards practical quantum variational algorithms, *Phys. Rev. A* **92**, 042303 (2015).
 - [5] C. Cade, L. Mineh, A. Montanaro, and S. Stanisic, Strategies for solving the fermi-hubbard model on near-term quantum computers, *Phys. Rev. B* **102**, 235122 (2020).
 - [6] E. R. Anschuetz and B. T. Kiani, Quantum variational algorithms are swamped with traps, *Nature Communications* **13**, 10.1038/s41467-022-35364-5 (2022).
 - [7] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, Barren plateaus in quantum neural network training landscapes, *Nature Communications* **9**, 10.1038/s41467-018-07090-4 (2018).
 - [8] H. R. Grimsley, S. E. Economou, E. Barnes, and N. J. Mayhall, An adaptive variational algorithm for exact molecular simulations on a quantum computer, *Nature Communications* **10**, 10.1038/s41467-019-10988-2 (2019).
 - [9] J. Leng, E. Hickman, J. Li, and X. Wu, Quantum hamiltonian descent (2023), arXiv:2303.01471 [quant-ph].
 - [10] M. Ma, R. Jaiswal, and M. Y. Niu, Mosaic: Scalable probabilistic error cancellation via variational blockwise noise aggregation (2026), arXiv:2603.26063 [quant-ph].
 - [11] F. Ticozzi and L. Viola, Quantum markovian subsystems: Invariance, attractivity, and control (2008), arXiv:0705.1372 [quant-ph].
 - [12] P. Pfeuty, The one-dimensional Ising model with a transverse field, *Annals of Physics* **57**, 79 (1970).
 - [13] P. Calabrese and J. Cardy, Entanglement entropy and conformal field theory, *Journal of Physics A: Mathematical and Theoretical* **42**, 504005 (2009).
 - [14] B. D. Mafalda Ramoa *et al.*, CEO-ADAPT-VQE, <https://github.com/mafaldaramoa/ceo-adapt-vqe> (2023).
 - [15] A. Y. Kitaev, Quantum measurements and the abelian stabilizer problem (1995), arXiv:quant-ph/9511026 [quant-ph].
 - [16] L. Lin and Y. Tong, Near-optimal ground state preparation, *Quantum* **4**, 372 (2020).
 - [17] Y. Dong, L. Lin, and Y. Tong, Ground-state preparation and energy estimation on early fault-tolerant quantum computers via quantum eigenvalue transformation of unitary matrices, *PRX Quantum* **3**, 10.1103/prxquantum.3.040305 (2022).
 - [18] M. Ma and J. Rimika, MoSAIC, <https://github.com/maya-maye/MoSAIC> (2025).
 - [19] M. Larocca, P. Czarnik, K. Sharma, G. Muraleedharan, P. J. Coles, and M. Cerezo, Diagnosing barren plateaus with tools from quantum optimal control, *Quantum* **6**, 824 (2022).
 - [20] W. W. Ho and T. H. Hsieh, Efficient variational simulation of non-trivial quantum states, *SciPost Physics* **6**, 10.21468/scipostphys.6.3.029 (2019).
 - [21] V. Jurdjevic and H. J. Sussmann, Control systems on lie groups, *Journal of Differential Equations* **12**, 313 (1972).

Appendix A: Different ansatz families tested

To benchmark the VHD construction, we compare it against two standard variational ansatz families for the transverse field Ising model: a problem specific ansatz built directly from the non-commuting terms of the target Hamiltonian, and a hardware-efficient ansatz built from layers of single-qubit rotations and entangling gates. In each case, we also consider a kinetic augmented version obtained by inserting the same kinetic generator

$$K = \sum_i Y_i$$

used in the VHD construction.

1. Problem specific ansatz

A natural problem-specific ansatz for the TFIM is obtained by alternating the two non-commuting pieces of

the target Hamiltonian [20]. For circuit depth p , we take

$$U_{\text{ps}}(\boldsymbol{\theta}^{ZZ}, \boldsymbol{\theta}^X) = \prod_{m=1}^p \left(\prod_{\langle q, q' \rangle} e^{-i\theta_{m,q,q'}^{ZZ} Z_q Z_{q'}} \right) \left(\prod_q e^{-i\theta_{m,q}^X X_q} \right) \quad (\text{A1})$$

This ansatz is tailored to the structure of the target Hamiltonian and may be viewed as a standard Hamiltonian variational ansatz for the TFIM.

To convert this to a VHD circuit, we add additional kinetic drive layers of the form

$$U_{\text{ps+K}}(\boldsymbol{\theta}^{ZZ}, \boldsymbol{\theta}^X, \Phi) = \prod_{m=1}^p \left(\prod_{\langle q, q' \rangle} e^{-i\theta_{m,q,q'}^{ZZ} Z_q Z_{q'}} \right) \left(\prod_q e^{-i\theta_{m,q}^X X_q} \right) e^{-i\Phi_m K}. \quad (\text{A2})$$

Most results presented in the main text belong to this family of VHD ansatz.

2. Hardware efficient ansatz

As a more architecture-oriented baseline, we also compare against a hardware-efficient ansatz composed of layers of single-qubit rotations followed by entangling gates arranged according to the device connectivity. In the implementation studied here, we use layers of single qubit X and Z rotations together with nearest neighbor entangling gates, giving

$$U_{\text{HEA}}(\boldsymbol{\vartheta}, \boldsymbol{\varphi}) = \prod_{m=1}^p \left(\prod_{\langle q, q' \rangle} \text{CZ}_{q,q'} \right) \left(\prod_{q=1}^N e^{-i\vartheta_m^q X_q} e^{-i\varphi_m^q Z_q} \right).$$

Here p denotes the circuit depth, and the angles ϑ_m^q and φ_m^q are treated as independent variational parameters.

To assess whether the kinetic drive is beneficial even in a more generic ansatz family, we also consider the kinetic augmented hardware efficient form

$$U_{\text{HEA+K}}(\boldsymbol{\vartheta}, \boldsymbol{\varphi}, \Phi) = \prod_{m=1}^p \left(\prod_{\langle q, q' \rangle} \text{CZ}_{q,q'} \right) \left(\prod_{q=1}^N e^{-i\vartheta_m^q X_q} e^{-i\varphi_m^q Z_q} \right) e^{-i\Phi_m K}. \quad (\text{A3})$$

Comparing Eqs. (??) and (A3) allows us to isolate the effect of the kinetic layers within a hardware-efficient circuit family that is not explicitly adapted to the structure of the TFIM Hamiltonian.

Here, we show additional results for the hardware-efficient family, which demonstrate equally good or even better results than the problem-specific case for the hardware-efficient ansatz.

Since hardware-efficient ansätze are often associated with difficult trainability, these results suggest that the VHD modification can improve optimization performance even in ansatz families where barren-plateau-related issues are a concern.

Appendix B: No-go theorems for QHD-based state preparation

The fundamental difficulty of variational state preparation does not arise from spurious local minima of the target energy on the full projective Hilbert space, the Rayleigh quotient of a self-adjoint Hamiltonian has the correct global variational structure, with ground state rays as its global minima and eigenspaces as its critical manifolds. The difficulty arises from the restriction to a physically realizable family of controls or ansatz states. The propositions below make this precise in the QHD setting, establishing two obstructions to treating coherent target-plus-kinetic dynamics as a universal descent mechanism.

Let K be a self-adjoint operator playing the role of a kinetic or mixer term, and consider the coherent control Hamiltonian

$$H(t) = a(t)H_T + b(t)K, \quad (\text{B1})$$

where $a, b \in L^\infty([0, T])$ are real-valued control functions.

Proposition 2 (Target energy is not a universal Lyapunov function). *Let $\mathcal{H}$ be finite dimensional, let H_T and K be self-adjoint operators on $\mathcal{H}$ with $[K, H_T] \neq 0$, and let $|\psi_t\rangle$ satisfy*

$$i\partial_t |\psi_t\rangle = (a(t)H_T + b(t)K)|\psi_t\rangle. \quad (\text{B2})$$

Define $E(t) := \langle \psi_t | H_T | \psi_t \rangle$. Then

$$\dot{E}(t) = i b(t) \langle \psi_t | [K, H_T] | \psi_t \rangle. \quad (\text{B3})$$

In particular, if $b(t_*) \neq 0$ at some time t_* , then there exist normalized states $|\psi_+\rangle$ and $|\psi_-\rangle$ such that

$$\dot{E}(t_*)|_{\psi_t=\psi_+} > 0, \quad \dot{E}(t_*)|_{\psi_t=\psi_-} < 0.$$

Hence $E(t)$ is not a Lyapunov function for the coherent dynamics.

Proof. Differentiating,

$$\dot{E}(t) = \langle \partial_t \psi_t | H_T | \psi_t \rangle + \langle \psi_t | H_T | \partial_t \psi_t \rangle \quad (\text{B4})$$

$$= i \langle \psi_t | [a(t)H_T + b(t)K, H_T] | \psi_t \rangle \quad (\text{B5})$$

$$= i b(t) \langle \psi_t | [K, H_T] | \psi_t \rangle, \quad (\text{B6})$$

using $[H_T, H_T] = 0$.

Let $A := i[K, H_T]$, which is Hermitian and nonzero. Since $\text{tr}(A) = i \text{tr}([K, H_T]) = 0$, A has both positive and negative eigenvalues. Choosing normalized eigenvectors $|\psi_+\rangle$ and $|\psi_-\rangle$ corresponding to positive and negative eigenvalues yields the claim. $\square$

The previous proposition rules out the target energy as a Lyapunov candidate. One might ask whether a different monotone function of the energy, such as E^2 or $\exp(E)$, could restore monotonicity. The next proposition shows this is impossible.

Proposition 3 (No universal Lyapunov function from target energy). *Under the assumptions of Proposition 2, assume additionally that there exists a time t_* with $b(t_*) \neq 0$. Let $L(\psi) = f(\langle \psi | H_T | \psi \rangle)$ where $f : \mathbb{R} \rightarrow \mathbb{R}$ is strictly monotone and differentiable. Then L cannot satisfy $\dot{L}(t_*) \leq 0$ for all normalized states, and hence cannot be a universal Lyapunov function for the dynamics.*

Proof. Let $A := i[K, H_T] \neq 0$. In an eigenbasis $\{|j\rangle\}$ of H_T with eigenvalues $\{E_j\}$,

$$A_{jk} = i(E_k - E_j)K_{jk}.$$

Since $A \neq 0$, there exist $j \neq k$ such that $A_{jk} \neq 0$ and $E_j \neq E_k$.

For $0 < p < 1$ and $\phi \in \mathbb{R}$, define

$$|\psi_\phi\rangle = \sqrt{p}|j\rangle + e^{i\phi}\sqrt{1-p}|k\rangle.$$

Then

$$\langle \psi_\phi | H_T | \psi_\phi \rangle = pE_j + (1-p)E_k$$

is independent of ϕ , while

$$\langle \psi_\phi | A | \psi_\phi \rangle = 2\sqrt{p(1-p)} \operatorname{Re}(e^{-i\phi} A_{jk}).$$

Choosing $\phi_\pm = \arg(A_{jk})$ and $\arg(A_{jk}) + \pi$ yields two states with identical energy but opposite signs of $\dot{L}(t_*)$.

Since f is strictly monotone, f' does not vanish identically on (E_j, E_k) , so p can be chosen such that $f'(E) \neq 0$. Then

$$\dot{L}(t_*) = f'(E)\dot{E}(t_*)$$

inherits the sign indefiniteness of $\dot{E}$, and cannot be non-positive for all states. $\square$

Proposition 4 (No universal asymptotic coherent preparation). *Let $U(t)$ be a family of unitary operators on $\mathcal{H}$, and suppose there exists a normalized state $|g\rangle$ such that for every normalized $|\psi\rangle$ in a set $\mathcal{S}$,*

$$U(t)|\psi\rangle \rightarrow e^{i\vartheta_\psi(t)}|g\rangle \quad \text{in norm as } t \rightarrow \infty.$$

Then every pair of states in $\mathcal{S}$ is collinear.

Proof. Unitarity gives

$$\langle \psi | \phi \rangle = \langle U(t)\psi, U(t)\phi \rangle \quad \text{for all } t.$$

By assumption,

$$\|U(t)\psi - e^{i\vartheta_\psi(t)}|g\rangle\| \rightarrow 0, \quad \|U(t)\phi - e^{i\vartheta_\phi(t)}|g\rangle\| \rightarrow 0.$$

Hence

$$|\langle \psi | \phi \rangle - e^{i(\vartheta_\phi(t) - \vartheta_\psi(t))}| \rightarrow 0.$$

Since the second term has unit modulus for all t , it follows that $|\langle \psi | \phi \rangle| = 1$. Equality in the Cauchy–Schwarz inequality implies that $|\psi\rangle$ and $|\phi\rangle$ are collinear. $\square$

Together, these three propositions clarify why QHD cannot be treated as a universal descent mechanism for quantum state preparation. Proposition 2 shows that the target energy cannot serve as a Lyapunov function for coherent dynamics whenever $[K, H_T] \neq 0$; Proposition 3 closes the obvious escape route by ruling out any monotone reparameterization of the energy; and Proposition 4 establishes the stronger global obstruction that unitarity itself prevents asymptotic preparation from any nontrivial set of initial states. These results are closely related in spirit to known limitations of closed quantum Lyapunov control: energy-based Lyapunov functions do not generically yield monotone descent for unitary dynamics, and pure-state stabilization fundamentally requires dissipative or measurement-based resources.

Instead of treating the continuous-time dynamics as a literal convergence mechanism, we go ahead and use the same control structure as the architectural blueprint for a variational ansatz, and ask whether the resulting energy landscape on the variational manifold acquires provably favorable structure. This is the question taken up in Appendix C.

Appendix C: VHD yields benign optimization geometry in the quadratic model

We now show that the no-go results of Appendix B do not preclude the VHD construction from producing provably favorable optimization landscapes. In a structured quadratic sector, the induced variational manifold inherits a globally benign optimization geometry that can be established rigorously. We proceed in three steps. The depth one case already exhibits the key features sharply (Theorem 5); the depth two analysis shows these features persist and in fact strengthen when an intermediate target layer is added (Theorem 6); and an intrinsic description at arbitrary depth explains why increasing circuit depth introduces redundant parameterizations thus not adding to the landscape complexity (Theorem 7). We close by showing that the no-go obstruction persists within the Gaussian sector, and explaining why this does not contradict the positive preparation results.

Depth one VHD ansatz

Theorem 5 (Global convergence on the depth one Gaussian VHD manifold). *Let*

$$H_\omega = \frac{1}{2}(P^2 + \omega^2 X^2), \quad K = \frac{1}{2}(XP + PX), \quad [X, P] = i,$$

with $\omega > 0$, and let $|0\rangle$ denote the ground state of $H_1 = \frac{1}{2}(P^2 + X^2)$. Consider the depth one VHD family

$$|\psi(\theta, \phi)\rangle = e^{-i\theta H_\omega} e^{-i\phi K} |0\rangle,$$

and define the target energy $E(\theta, \phi) := \langle \psi(\theta, \phi) | H_\omega | \psi(\theta, \phi) \rangle$. Then:

1. $E(\theta, \phi)$ is independent of θ and is given by

$$E(\phi) = \frac{1}{4}(e^{-2\phi} + \omega^2 e^{2\phi}). \quad (C1)$$

2. $E(\phi)$ is strictly convex on $\mathbb{R}$, with unique global minimizer

$$\phi_\star = -\frac{1}{2} \log \omega, \quad (C2)$$

at which $E(\phi_\star) = \omega/2$, the exact ground state energy of H_ω .

3. The state $e^{-i\phi_\star K} |0\rangle$ is the ground state of H_ω up to an overall phase.
4. The gradient flow $\dot{\phi} = -E'(\phi)$ converges globally to $\phi_\star$ for every initial $\phi(0) \in \mathbb{R}$.

Proof. Since H_ω commutes with its own exponential, $E(\theta, \phi) = \langle 0 | e^{i\phi K} H_\omega e^{-i\phi K} | 0 \rangle$, which is independent of θ . The squeezing generator K acts in the Heisenberg picture as

$$e^{i\phi K} X e^{-i\phi K} = e^\phi X, \quad e^{i\phi K} P e^{-i\phi K} = e^{-\phi} P,$$

so

$$e^{i\phi K} H_\omega e^{-i\phi K} = \frac{1}{2}(e^{-2\phi} P^2 + \omega^2 e^{2\phi} X^2).$$

Taking the expectation in $|0\rangle$ and using $\langle 0 | X^2 | 0 \rangle = \langle 0 | P^2 | 0 \rangle = \frac{1}{2}$, which follows from the standard harmonic oscillator vacuum ($[X, P] = i$, $H_1 |0\rangle = \frac{1}{2} |0\rangle$), gives (C1). Differentiating, $E''(\phi) = e^{-2\phi} + \omega^2 e^{2\phi} > 0$, so E is strictly convex. The unique critical point $\phi_\star$ satisfies $e^{-2\phi_\star} = \omega^2 e^{2\phi_\star}$, giving $\phi_\star = -\frac{1}{2} \log \omega$ and $E(\phi_\star) = \omega/2$. At $\phi_\star$, the squeezed vacuum has covariances $\langle X^2 \rangle = 1/(2\omega)$ and $\langle P^2 \rangle = \omega/2$, which are exactly those of the ground state of H_ω ; since a centered pure Gaussian state is uniquely determined by its covariance matrix, this identifies $e^{-i\phi_\star K} |0\rangle$ as the ground state up to phase. Global convergence of the gradient flow follows from strict convexity and coercivity of E on $\mathbb{R}$: $E'(\phi) < 0$ for $\phi < \phi_\star$ and $E'(\phi) > 0$ for $\phi > \phi_\star$, so every trajectory is attracted to $\phi_\star$. $\square$

Depth two VHD ansatz

Before turning to depth two, we note a structural redundancy that simplifies the analysis. For any circuit of the form $|\psi(\theta)\rangle = e^{-i\theta H_\omega} |\chi\rangle$, the target energy satisfies $\langle \psi(\theta) | H_\omega | \psi(\theta) \rangle = \langle \chi | H_\omega | \chi \rangle$, since H_ω commutes

with its own exponential. Consequently, an outermost target layer is invisible to the energy objective, and the depth two ansatz $e^{-i\theta_2 H_\omega} e^{-i\phi_2 K} e^{-i\theta_1 H_\omega} e^{-i\phi_1 K} |0\rangle$ has an energy that is entirely independent of θ_2 . Without loss of generality we therefore fix the outer target angle and study the reduced circuit $|\psi(\phi_1, \phi_2)\rangle = e^{-i\phi_2 K} e^{-i\theta H_\omega} e^{-i\phi_1 K} |0\rangle$.

Theorem 6 (Depth two Gaussian VHD landscape). *Under the same setup as Theorem 5, fix $\theta \in \mathbb{R}$ and set $c := \cos(\omega\theta)$, $s := \sin(\omega\theta)$, $u := \phi_1 + \phi_2$, $v := \phi_1 - \phi_2$. Define $E(\phi_1, \phi_2) := \langle \psi(\phi_1, \phi_2) | H_\omega | \psi(\phi_1, \phi_2) \rangle$. Then:*

1. The energy separates exactly as

$$E(\phi_1, \phi_2) = c^2 E_{\text{single}}(u) + s^2 E_{\text{single}}(v),$$

$$E_{\text{single}}(x) := \frac{1}{4}(\omega^2 e^{2x} + e^{-2x}). \quad (C3)$$

2. In the generic case $c \neq 0$ and $s \neq 0$, E is globally strongly convex with

$$\nabla^2 E(\phi_1, \phi_2) \succeq 4\omega \min(c^2, s^2) I_2 \quad \forall (\phi_1, \phi_2) \in \mathbb{R}^2. \quad (C4)$$

The unique global minimizer is

$$\phi_1^* = -\frac{1}{2} \log \omega, \quad \phi_2^* = 0, \quad (C5)$$

at which $E(\phi_1^*, \phi_2^*) = \omega/2$.

3. In the resonant cases $c = 0$ or $s = 0$, the minimum value remains $\omega/2$, but the minimizer becomes an affine line ($\phi_1 + \phi_2 = -\frac{1}{2} \log \omega$ if $s = 0$; $\phi_1 - \phi_2 = -\frac{1}{2} \log \omega$ if $c = 0$).

4. In the generic case, the Euclidean gradient flow converges globally and exponentially:

$$\|\Phi(t) - \Phi_\star\| \leq e^{-mt} \|\Phi(0) - \Phi_\star\|,$$

$$m := 4\omega \min(c^2, s^2). \quad (C6)$$

Proof. We track the covariance matrix $\Sigma_{ij} := \frac{1}{2} \langle R_i R_j + R_j R_i \rangle$, $R = (X, P)^T$, through each layer. The reference vacuum has $\Sigma_0 = \frac{1}{2} I_2$. After the first kinetic layer,

$$\Sigma_1 = \frac{1}{2} \begin{pmatrix} e^{2\phi_1} & 0 \\ 0 & e^{-2\phi_1} \end{pmatrix}.$$

The target evolution $e^{-i\theta H_\omega}$ acts by the symplectic rotation $M_\theta = \begin{pmatrix} c & s/\omega \\ -\omega s & c \end{pmatrix}$, giving $\Sigma_2 = M_\theta \Sigma_1 M_\theta^T$ with diagonal entries

$$(\Sigma_2)_{11} = \frac{1}{2} \left(c^2 e^{2\phi_1} + \frac{s^2}{\omega^2} e^{-2\phi_1} \right),$$

$$(\Sigma_2)_{22} = \frac{1}{2} \left(\omega^2 s^2 e^{2\phi_1} + c^2 e^{-2\phi_1} \right).$$

After the second kinetic layer, $(\Sigma_3)_{11} = e^{2\phi_2} (\Sigma_2)_{11}$ and $(\Sigma_3)_{22} = e^{-2\phi_2} (\Sigma_2)_{22}$. Since $E = \frac{\omega^2}{2} (\Sigma_3)_{11} + \frac{1}{2} (\Sigma_3)_{22}$, substituting gives (C3) with $u = \phi_1 + \phi_2$ and $v = \phi_1 - \phi_2$.

Since E_{single} is strictly convex in x , the energy E is a strictly positive weighted sum of two convex functions in the independent variables u and v , and its unique minimizer in the generic case is $u_* = v_* = -\frac{1}{2} \log \omega$, giving (C5). For the Hessian, differentiating the separated form in the original variables yields

$$\nabla^2 E = c^2 E''_{\text{single}}(u) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + s^2 E''_{\text{single}}(v) \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

These two matrices have eigenvectors $(1, 1)^T$ and $(1, -1)^T$ with eigenvalues $2c^2 E''_{\text{single}}(u)$ and $2s^2 E''_{\text{single}}(v)$ respectively. By AM-GM, $E''_{\text{single}}(x) = \omega^2 e^{2x} + e^{-2x} \geq 2\omega$ for all x , giving (C4). Exponential convergence of gradient flow then follows from standard strongly convex potential theory. $\square$

The minimizer (C5) carries a notable physical interpretation: the nontrivial kinetic action is entirely front-loaded into the first layer ($\phi_1^* = -\frac{1}{2} \log \omega$), while the second kinetic layer switches off at optimality ($\phi_2^* = 0$). Adding depth therefore does not create landscape complexity; global strong convexity is preserved while naturally favoring a structure in which earlier layers carry stronger kinetic action - precisely the QHD-inspired design principle!

Arbitrary depth: intrinsic description

While explicit covariance calculations become cumbersome at larger depth, the quadratic setting admits a cleaner intrinsic description. The key observation is that any finite depth VHD circuit generated by H_ω and K remains within the manifold of centered pure Gaussian states, and on that manifold the target energy depends on only a single intrinsic coordinate. This yields a depth-independent result explaining why additional layers only introduce coordinate redundancy.

For any depth $p \geq 1$, consider the alternating circuit

$$|\psi(\theta, \phi)\rangle = \prod_{j=1}^p e^{-i\theta_j H_\omega} e^{-i\phi_j K} |0\rangle. \quad (\text{C7})$$

Introduce the scaled quadratures $\tilde{X} := \sqrt{\omega} X$ and $\tilde{P} := P/\sqrt{\omega}$, in which $H_\omega = \frac{\omega}{2}(\tilde{X}^2 + \tilde{P}^2)$, and let $\tilde{\Sigma}$ denote the covariance matrix of the state in these coordinates.

Theorem 7 (Intrinsic Gaussian energy profile for arbitrary depth). *For every finite depth p and every parameter choice (θ, ϕ) , the state (C7) is a centered pure Gaussian state with $\det \tilde{\Sigma} = 1/4$. There exist an orientation angle $\alpha \in [0, \pi)$ and an intrinsic squeezing magnitude $r \geq 0$ such that*

$$\tilde{\Sigma} = \frac{1}{2} \mathcal{R}(\alpha) \begin{pmatrix} e^{2r} & 0 \\ 0 & e^{-2r} \end{pmatrix} \mathcal{R}(\alpha)^T, \quad (\text{C8})$$

and the target energy depends only on r ,

$$E(\theta, \phi) = \frac{\omega}{2} \cosh(2r). \quad (\text{C9})$$

Consequently, $E \geq \omega/2$ with equality if and only if $r = 0$, i.e., the state is the ground state of H_ω . The orientation angle α plays no role in the energy.

Proof. Since H_ω and K are both quadratic, the associated unitaries preserve centered pure Gaussian states, so every state of the form (C7) is centered and pure Gaussian. In the scaled quadratures, $e^{-i\theta H_\omega}$ acts by the rotation $\mathcal{R}(\omega\theta)$ and $e^{-i\phi K}$ acts by the diagonal squeeze $\text{diag}(e^\phi, e^{-\phi})$, both of which are real symplectic maps with determinant one. Hence $\det \tilde{\Sigma} = \det \tilde{\Sigma}_0 = 1/4$ is preserved throughout. Any positive symmetric 2×2 matrix with determinant $1/4$ can be parameterized as (C8) by the standard one-mode pure Gaussian decomposition. Since $H_\omega = \frac{\omega}{2}(\tilde{X}^2 + \tilde{P}^2)$, the energy of any centered state is $E = \frac{\omega}{2} \text{tr} \tilde{\Sigma}$. By orthogonal invariance of the trace, $\text{tr} \tilde{\Sigma} = \frac{1}{2}(e^{2r} + e^{-2r}) = \cosh(2r)$, giving (C9). The remaining claims follow from $\cosh(2r) \geq 1$ with equality iff $r = 0$. $\square$

Theorem 7 identifies the correct positive statement at arbitrary depth. The energy excess above the ground state is

$$E - \frac{\omega}{2} = \omega \sinh^2 r, \quad (\text{C10})$$

so the energy excess is an exact monotone function of the intrinsic squeezing magnitude r . Additional layers introduce flat directions and redundant parameterizations of the same low-dimensional manifold without creating new intrinsic minima. It is important, however, not to over-generalize this observation: the strong redundancy established here is a special feature of the one-mode quadratic sector. Once one allows nonquadratic generators, non-Gaussian reference states, or interacting many-body targets, the circuit generally leaves the Gaussian manifold and these exact flat directions can be lifted.

The no-go obstruction within the Gaussian sector

It is natural to ask whether the continuous-time QHD dynamics itself converges within the Gaussian sector, or whether the no-go obstruction of Appendix B is already visible here. In fact, the obstruction persists even in this solvable setting. Writing the target energy as $E = \frac{1}{2} \Sigma_{22} + \frac{\omega^2}{2} \Sigma_{11}$ and computing $\dot{E}$ under $H(t) = a(t)H_\omega + b(t)K$, one finds

$$\dot{E} = \omega b(t) \sinh(2r) \cos(2\alpha), \quad (\text{C11})$$

where r and α are the intrinsic squeezing magnitude and orientation of the state. For any time t_* with $b(t_*) \neq 0$ and any $r > 0$, the two states with the same squeezing

magnitude but orientations $\alpha = 0$ and $\alpha = \pi/2$ experience equal and opposite energy derivatives under the same schedule. The no-go obstruction is therefore fully present within the Gaussian manifold: there is no state-independent schedule that guarantees monotonic energy descent.

What the Gaussian case possesses, and the general case does not, is controllability. The generators H_ω and K generate $\mathfrak{sp}(2, \mathbb{R})$ as a Lie algebra, and the connected symplectic group $Sp(2, \mathbb{R})$ acts transitively on centered pure Gaussian states [21]. Consequently, any centered pure Gaussian state can be steered to any other by a finite concatenation of evolutions generated by H_ω and K . In particular, for any initial centered pure Gaussian state, there exists a finite-time schedule $(a(t), b(t))$ steering it exactly to the ground state of H_ω .

This controllability explains why the no-go theorem is

consistent with the positive preparation results above. The no-go statement is uniform over all initial states, whereas the Gaussian example concerns a specific reference state i.e., the vacuum of H_1 , from which the ground state of H_ω lies on the same squeezed orbit and is therefore coherently reachable. The Gaussian sector is special not because the obstruction is absent, but because the target is reachable from the chosen reference state and the variational landscape restricted to that orbit remains globally convex.

VHD turns this observation into a practical strategy: rather than constructing the optimal coherent schedule analytically, it parameterizes the same structured family of evolutions and recovers the correct one through gradient optimization on the provably convex restricted landscape.